



**Verified Carbon
Standard**

INSTALLATION OF HIGH EFFICIENCY COOKSTOVES IN INDIA BY AGS CARBON



AGS Carbon Advisory

Document Prepared by AGS Carbon Advisory

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CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Sectoral Scope and Project Type	4
1.3	Project Eligibility	5
1.4	Project Design	6
1.5	Project Proponent	9
1.6	Other Entities Involved in the Project	9
1.7	Ownership.....	10
1.8	Project Start Date	10
1.9	Project Crediting Period	10
1.10	Project Scale and Estimated GHG Emission Reductions or Removals	10
1.11	Description of the Project Activity	11
1.12	Project Location	12
1.13	Conditions Prior to Project Initiation	13
1.14	Compliance with Laws, Statutes and Other Regulatory Frameworks.....	13
1.15	Participation under Other GHG Programs	13
1.16	Other Forms of Credit.....	13
1.17	Sustainable Development Contributions	14
1.18	Additional Information Relevant to the Project	15
2	SAFEGUARDS	15
2.1	No Net Harm	15
2.2	Local Stakeholder Consultation	15
2.3	Environmental Impact	19
2.4	Public Comments	19
2.5	AFOLU-Specific Safeguards	19
3	APPLICATION OF METHODOLOGY.....	20
3.1	Title and Reference of Methodology	20
3.2	Applicability of Methodology	20
3.3	Project Boundary	23
3.4	Baseline Scenario	24

3.5	Additionality	25
4	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	26
4.1	Baseline Emissions	26
4.2	Project Emissions	26
4.3	Leakage.....	26
4.4	Net GHG Emission Reductions and Removals	26
5	MONITORING	30
5.1	Data and Parameters Available at Validation	30
5.2	Data and Parameters Monitored.....	34
5.3	Monitoring Plan.....	37

1 PROJECT DETAILS

1.1 Summary Description of the Project

The purpose of this Grouped project (GP) “Installation of high efficiency cookstoves in India by AGS Carbon” is to reduce the rate of deforestation in the remote areas along with reduction of the indoor air pollution. The GP will reduce the need for women to collect firewood for longer time. The GP will distribute 75,000 ICS (improved cookstoves) to the beneficiaries.

Approximately, 3.8 million people every year die prematurely from illness attributable to the household air pollution caused by the inefficient use of solid fuels and kerosene for cooking¹. The efficient cookstove has a better combustion process due to which indoor air pollution along with GHG emissions is significantly reduced compared to traditional three stone cooking method. The beneficiaries will be the low-income households in rural Maharashtra, India. The geographical location of the project is Maharashtra, India.

The baseline scenario is continuous use of firewood using three-stone fired stove to meet the energy demand for cooking purposes with traditional method of open fire or three stone cookstove. The project proponent of the project is “AGS Carbon Advisory”. Each household will receive one ICS and the project will target only the end users with three stone-fired stove using firewood.

The group project is estimated to offset an annual average of 333,938 tCO₂ emission reductions and a total of 2,337,567 tCO₂ emission reductions over the crediting period of 7 years (6 June 2022 to 5 June 2029).

1.2 Sectoral Scope and Project Type

Applied Methodology – VMR0006: Methodology for installation of high efficiency firewood cookstove; Version: 1.1;²

Sectoral scope: 03 Energy Demand

Type II- Energy Efficiency Improvement project

The project is a Grouped Project

¹ <https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health>

² <https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf>

1.3 Project Eligibility

Grouped Project and Project Activity Instances

The GP falls within the scope of the VCS Program as it is in line with the VCS Standard v 4.3 section 2.1.1. The Grouped project includes reduction of Carbon Dioxide emissions which is one of the six Kyoto Protocol greenhouse gases included under the scope of VCS program. And the Grouped Project is supported by a methodology approved under an approved GHG Program by VCS. The project does not involve use of Ozone Depleting Substances, Jurisdictional or nested REDD+ programs and doesn't use a VCS methodology.

Further, the project has not generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction.

The project does not fall under the category of excluded projects in Table 1 of the VCS Standard v4.3 and is therefore eligible under the scope of the VCS Program:

Excluded Activity	Applicability
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	N/A: The GP reduces GHG emissions due to energy efficiency measures by introducing improved cookstoves
Grid-connected electricity generation using hydro-power plants/units	N/A: The GP reduces GHG emissions due to energy efficiency measures by introducing improved cookstoves and does not involve generation of electricity.
Grid-connected electricity generation using wind, geothermal, or solar power plants/units	N/A: The GP reduces GHG emissions due to energy efficiency measures by introducing improved cookstoves and does not involve generation of electricity.
Utilization of recovered waste heat for, inter alia combined cycle electricity generation and the provision of heat for residential, commercial or industrial use	N/A: The GP reduces GHG emissions due to energy efficiency measures by introducing improved cookstoves and does not involve generation of heat for producing electricity.
Generation of electricity and/or thermal energy using biomass. This does not include efficiency improvements in thermal applications (e.g, cookstoves)	N/A: The GP reduces GHG emissions due to energy efficiency measures by introducing improved cookstoves and does not involve generation of heat for producing electricity.

Generation of electricity and/or thermal energy using fossil fuels, including activities that involve switching from a higher carbon content fuel to a lower carbon content fuel	N/A: The GP activities do not include the generation of electricity and/or thermal energy using fossil fuels, including activities that involve switching from a higher carbon content fuel to a lower carbon content fuel.
Replacement of electric lighting with more energy efficient electric lighting such as the replacement of incandescent electrical bulbs with CFLs or LEDs	N/A: The GP activities do not include the replacement of electric lighting with more energy efficient lighting.
Installation and/or replacement of electricity transmission lines and/or energy efficient transformers	N/A: The GP activities do not include the installation and/or replacement of electricity transmission lines and/or energy efficient transformers.

Table 1 Excluded Project Activities under VCS

1.4 Project Design

The project is a grouped project.

Eligibility Criteria

For the inclusion of new project instances (PIs), the project proponent shall ensure that it meets the eligibility criteria set out in clause 3.5.15 and 3.5.16 of the VCS Standard v4.3.

S. No.	Eligibility criteria for the inclusion of new project activity instances	How the new instances will comply	How the PI is complying and Evidences
1	Meet the applicability conditions set out in the methodology applied to the project	New project activity instances ICS (Improved Cookstoves) will meet the applicability conditions set out in Section 3.2 of this document.	Please see section 3.2 of this document for details.
2	Occur within one of the designated geographic areas specified in the project description.	New project activity instances will be located within the geographic boundaries of Maharashtra, India. The new Project Instances geographic	The Project Instances in first Project Activity are located in rural regions of Maharashtra state in India. Evidence: a) Project database (sales

		location will be defined in the monitoring reports.	record) with contact details of HHs.
3	Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only).	Each PI within this GP shall be eligible for crediting from its start date through to the end of the project crediting period (only).	The project is eligible for crediting from start date, which is 6 June 2022. Evidence: Distribution Database shared with VVB
4	Have a start date that is the same as or later than the grouped project start date.	Only PIs having a start date that is the same as or later than the GP start date will be included in the GP. The start date of GP is the date of distribution of first unit as per section 1.8 of this document. And subsequent additions will have start date after GP start date only.	The start date of project is 6 June 2022 (start date of distribution of first ICS (PI) under the Grouped project) which is same as the start date of the Grouped Project. Evidence: Distribution database
5	The specifications of technologies or measures including the level and type of service performance specifications including compliance with testing/certifications	Project instances under this project will consist of the distribution of ICS with a thermal efficiency of at least 25% efficiency as per water boiling test to household users cooking with non-renewable biomass in the baseline scenario. The project instances will involve replacement of conventional firewood cook-stoves with ICS as defined in the VCS PD. Stove types replaced and implemented is defined in the VCS PD, hence involve efficiency improvements in the thermal applications of non-renewable biomass appliances as per VMR0006	The PI cookstoves have higher efficiency and better emission parameters compared to the baseline scenario. Evidence: Manufacturer's specification/distribution records

6	VER ownership	End users receiving ICSs under the specific project instances contractually cede their rights to claim and own emission reductions under the VCS to the PP of the project.	A default distribution agreement for end users including the provision that emission reductions generated by the improved stove are owned by the PP will be provided for each household.
7	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	The new PIs will be installed within Maharashtra and new PIs shall be added subject to the same baseline scenario determined in Section 3.4 of this VPA-DD.	The PI 1 has the same baseline scenario as defined in section 3.4 of this document i.e. use of traditional three stone cookstove for cooking in different regions of Maharashtra. Evidence: A Baseline survey report which shall be shared with the VVB. This information will be stored during monitoring and in the database.
8	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	The PIs added to this GP shall comply with all the additionality conditions listed in section 3.5 and should be clearly defined in the Monitoring Reports.	The PI 1 is additional and is described in section 3.5 of this document. Step 1: Regulatory Surplus There is no mandated government programme or policy in host country of this project ensuring the distribution of new energy efficient cook stoves for each project activity instances. Step 2: Positive List Th project activity instances comply with positive list as it satisfies criterion 1 as the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG

			credits. Hence it is deemed additional. Evidence: End user interview and carbon title transfer form
9	The PP shall engage with local stakeholders during the project development and implementation processes.	The PP will conduct LSC before implementation of each Project activity i.e. before submission of Monitoring Reports for newly added PIs. The summary of LSC comments shall form part of the MR.	The PP conducted LSC in Nagpur, Maharashtra where stakeholders from the all regions where implementation of first Project activity will be implemented in Maharashtra state. LSC related documents have been submitted to VVB.

Table 2 Eligibility Criteria

1.5 Project Proponent

Organization name	AGS Carbon Advisory
Contact person	Mr. Abhishek Goyal
Title	Partner
Address	AGS Carbon Advisory, C-129, Lajpat Nagar-II, New Delhi-110024
Telephone	+91-9106557796
Email	info@agscarbon.com

1.6 Other Entities Involved in the Project

NA. AGS Carbon Advisory is the sole entity involved in the project.

Organization name	NA
Role in the project	NA
Contact person	NA
Title	NA

Address	NA
Telephone	NA
Email	NA

1.7 Ownership

The project ownership is with AGS Carbon Advisory. During the sale of ICS (Improved Cookstove), the participating household will sign a Declaration of Carbon Title Transfer to confirm that the ownership rights of the carbon assets generated from this project lie with the project proponent.

1.8 Project Start Date

6th June 2022 (date of sale of first ICS (improved cook stove) under the Grouped project and first Project Instance)

1.9 Project Crediting Period

Crediting Period: Renewable

Total number of years: 7 years

Start and End Date: 06/06/2022 to 05/06/2029

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	
Large project	Yes

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	397,252
Year 2	375,907
Year 3	354,704
Year 4	333,645
Year 5	312,733

Year 6	291,969
Year 7	271,356
Total estimated ERs	2,337,567
Total number of crediting years	7
Average annual ERs	333,938

1.11 Description of the Project Activity

The project involves distribution of fuel-efficient improved cookstoves (ICS) to replace the baseline three stone cookstoves in the households. The baseline 3-stone cookstove has an efficiency of 10% which is default as per VMR0006 methodology.



Figure 1 Baseline Cookstove (Three Stone)

The ICS to be deployed under this project substantially reduces fuel consumption and emissions for conducting cooking and water heating tasks in homes. The ICS improves the efficiency of combustion and thermal transfer to the pot compared with a traditional three-stone fire by incorporating a number of cutting-edge components, including a highly-insulated combustion chamber which provides a conducive environment for clean and efficient combustion of wood. It substantially reduces wood fuel consumption compared with a three-stone fire. Improved cook stove (ICS) disseminated under the project will be the Jumbo, a portable stove made of durable components. The initial model has a specified thermal efficiency of 38% and average lifespan of 6 years³. In case of the product becomes non-operation, replacement stoves will be provided to the end user. The single burner design ensures high efficiency. The power of the cookstove is 2.32 KW.

³ The lifetime may vary from individual product to product depending on usage handling and other physical factors. Additionally, there is an elaborate complaint registration system to assist in systematically tracking and resolving the customer complaints in timely manner.



Figure 2 Greenway Jumbo Cookstove (Project Scenario)

The Greenway Jumbo demonstrations in one of the villages are shown below.



Figure 3 Project Cookstove Demonstration

Other models of improved cookstove may also be offered under the project as long as they meet all the requirements of the methodology. Other models will also meet eligibility criterion of instances inclusion.

1.12 Project Location

The PIs will be implemented in rural areas of Maharashtra, India. The geographic coordinates of Maharashtra, India are 19° 39' 47.8080" N and 75° 18' 1.0548" E.



Figure 4 Map of Maharashtra, India

1.13 Conditions Prior to Project Initiation

Prior to the project activity these rural households were using inefficient traditional three stone cookstoves which is same as the baseline scenario. The baseline scenario has been described in detail in section 3.4

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no rules or regulations in the host country India or in the state of Maharashtra for implementation of the PIs. Though, the government of India promotes use of clean fuel e.g. LPG, however due to the initial capital cost and operating cost, the usage levels are low and most of the rural households still use traditional stoves for cooking.

1.15 Participation under Other GHG Programs

1.6 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has neither been registered, nor is seeking registration under any other GHG programs.

1.6 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs

1.16 Other Forms of Credit

1.6 Emissions Trading Programs and Other Binding Limits

This project is not part of any emissions trading program or any other mechanism that includes GHG allowance trading.

1.6 Other Forms of Environmental Credit

The project has not sought or will receive another form of GHG-related environmental credit.

1.17 Sustainable Development Contributions

India's sustainable development priorities are in line with the UN Sustainable Development Goals.

Environmental Sustainability

The project will help significantly reduce greenhouse gas emissions over its lifetime. (SDG 13)

The project will help reduce the use of non-renewable biomass from forests, thus assist in conserving existing forest stock and the protection of natural forest eco-systems and wildlife habitats.

Social Sustainability

Considerably less time need to be spent collecting wood fuel for the homes thereby reducing the work burden on rural families and presenting alternative opportunities for economic development. (SDG 5.4.1)

The amount of indoor pollutant from the burning of biomass in the homes will be reduced. Less carbon dioxide, carbon monoxide and particulates will be emitted due to the decrease in total biomass burned and an increase in the temperature of combustion⁴. (SDG 3.2.1)

Reduce instance of non-communicable respiratory disease, such as Chronic Obstructive Pulmonary Disease (COPD) among adult family members, particularly women and the elderly⁵. (SDG 3.4.1)

The stove provides a safer method for combusting biomass for cooking, helping to reduce burn injuries, especially for children. (SDG 3.2.2)

Economic Sustainability

The project will help develop a section of the local economy, in the distribution, local assembly, maintenance and monitoring activities.

⁴ <https://cleancooking.org/wp-content/uploads/2021/07/564-1.pdf>

⁵ <https://cleancooking.org/wp-content/uploads/2021/07/623-1.pdf>

Household expenditures on cooking fuel will be reduced through the use of the ICS.

Saved household labour can be diverted to more productive economic activities.

The project will create local employment opportunities in operational and management roles, as well as future assembly and/or manufacturing initiatives.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage

Commercially Sensitive Information

No information has been excluded from the public version of the project description.

Further Information

Not applicable

2 SAFEGUARDS

2.1 No Net Harm

No potential negative environmental or socio-economic impacts have been identified for the project.

2.2 Local Stakeholder Consultation

The local stakeholder consultation for the grouped project activity was conducted on 17/04/2022 at Mour Memorial Hall, Sitabardi, Nagpur, Maharashtra at 10 a.m. The stakeholder consultation was conducted at the grouped project activity level as the technology and its impact will be similar for all the project activities implemented in this grouped project activity.

The stakeholders from different categories were invited which represented the following groups:

- traditional stove users,
- local self-help groups,
- field assistants working in the project,

- local policy makers, and
- local NGOs.

The stakeholders were invited for the consultation through Notices and invitation letters to above categorized stakeholders. Detailed contact information such as phone number and email address of the representative of the project implementer were also made available in the Notices, invitation letters, for the stakeholders who could not physically attend the local stakeholder consultation. Due to the rising covid pandemic situation, online participation on a shared link was also encouraged.

A total of 32 participants attended the meeting. and of the people present each of them was allowed to participate in discussion.

AGS Carbon Advisory opened the meeting by thanking all the participants for taking out time from their busy schedules for this meeting, especially thanking the women for their participation.

AGS briefed the stakeholders on design of the project:

- a. The project will implement 75,000 stoves in Vidarbha region in the first instance and then will do more instances of 75,000 stoves as funding is raised.
- b. This program is totally financed by carbon finance. The stoves will be distributed free of cost by the NGOs involved in distribution and monitoring.
- c. The project will involve distribution of Greenway Grameen Jumbo stoves and other stove models. It will be ensured that whichever stove model is selected are eligible to be included in the programme and are of best quality..
- d. The main aim of the project is to avoid greenhouse gas emissions by reducing firewood consumption and protecting trees in the forest in this region. In addition, women and children will benefit from less smoke compared to traditional chullah. Women will also save time in collecting wood and cooking food as this stove cook's food much faster and needs much less wood.
- e. The process of VERRA validation, registration and issuance was explained and it was highlighted that there will be regular audits of the project.
- f. Respective NGOs working in different districts will set up helpline numbers to register any grievance from stakeholders and record any issues that users are facing with ICS. An online data base will be maintained to record if and when complaint from user is getting resolved.

- g. Since these stoves will have a direct impact on forests, it was suggested by stakeholders that we should inform forest department about this program. NGOs agreed to get the program flagged off by forest department.
- h. Two women participants who were given Greenway Grameen Jumbo Stoves for testing about 2 months ago were also present in the meeting. They gave very positive feedback on their experience. They said that the amount of wood required is really very low compared to traditional chullah and its completely smokeless stove. They can use small twigs that fall from trees and do not need big logs to wood to burn in this stove.
- i. It was suggested that the program should first prioritize very poor households and families affected by farmers suicide.
- j. Some of the questions raised during the consultation and their responses by AGS team are as follows:

S. No.	Question	AGS response
1.	How many times in a day can a user use this stove?	This stove can be used as many times as required in a day.
2	What is the weight of the stove?	It weighs 5 kg
3	What is the material used for making the handle?	It is made of special material that can withstand high temperatures and does not become hot.
4	How soon can we get the stove?	AGS replied that it plans to send first batch of 5000 stoves next month and then as further funding is raised more stoves will be sent for distribution.

Table 3: Q&A during LSC

The participants were provided with a feedback form to raise any concerns and comments for the project. Overall, the project received very warm reception and people were able to relate to the advantages from the Improved cookstove (ICS) and harmful effects of the baseline scenario.



Figure 5: Stakeholder Consultation

The table below lists few of the feedbacks received from the stakeholders. There were no negative comments from any section of the people to the project activity.

Name of Stakeholder	Question asked
Mr Amit Gaesail, Yavatmal	“Meeting is useful & healthy discussion” with “the project aiming to reduce smoke is a welcoming aspect.”
Mr. Vikram Wadhar, Yavatmal	“The meeting went well and the project is useful for reducing pollution and reduce time for wood collection”.
Ms. Neeta Sarkar, Ghatanji	“The meeting spoke about women’s health and environmental safety” while also adding that “the cost of LPG cylinder is high and this project aims to provide low-cost solution”
Mr. Akhil Ahamad, Gadchiroli	“The meeting was well organised” and “felt that the project was useful for conservation of environment and health”
Mr. Manag Dawale, Local NGO	“The process of implementation was clear and understood the project overall” and “the project helps reduce cost and time for cooking”

Ms. Shushma Shefali, Local NGO	“The project will help local woman to improve their health by reducing pollution from pre-project scenario”
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Table 4: Stakeholder Feedback

All the documents related to Local Stakeholder consultation- Notice, invitation letters, Attendance sheet, feedback forms will be shared with the validator.

AGS will ensure that regular monitoring will be conducted by the NGO staff to take feedback from the local stakeholders. There were no major comments from the stakeholders which could impact the Design of the Grouped project and hence no updates are required to the project Design.

For continuous input and grievance mechanism: The stakeholders were informed that the project has designed a continuous input and grievance mechanism to ensure all grievances are recorded and responded. The contact details and Location for maintaining a log book to record the query/complaints of the stakeholders was also shared:

Contact details for on-going communication:

AGS Carbon Advisory

C-129, Lajpat Nagar 2,

New Delhi - 110024

info@agscarbon.com

Dream Bahuudeshiya Sanstha

Trimurti Nagar Teosa,

Ta. Teosa, Dist. Amravati, Maharashtra – 444903

Contact details: 9420126091

Email ID: dreamngo.teosa@gmail.com

2.3 Environmental Impact

The Pls distributed under this Grouped project does not fall in the list of projects or activities requiring environmental impacts assessment.

2.4 Public Comments

No comments have been raised during the public comment period.

2.5 AFOLU-Specific Safeguards

Not applicable, since this is not an AFOLU project

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Applied Methodology– VMR 0006 Methodology for installation of high efficiency firewood cookstove; Version: 1.1; ⁶

Sectoral scope: 03 Energy Demand

The project has also referred to following documents:

Standard: VCS Standard, v4.3⁷

Tool 30: Calculation of the fraction of non-renewable biomass”; Version-03, EB 108, Annex 11

Guideline: General guidelines for SSC CDM methodologies Version 23.1⁸

3.2 Applicability of Methodology

Applicability criterion	How the project complies	Evidence
The project stove is a single pot or multi pot portable or an in-situ cookstove using only woody biomass; Additional requirement to demonstrate that the biomass used is solely renewable biomass in case project activities replacing baseline stoves using fossil fuel	The proposed instances will distribute high efficiency single-pot improved cook stove for thermal application, which will replace inefficient traditional cookstove leading to saving of non-renewable biomass. The ICS to be distributed under the project activity is Greenway Jumbo stoves/greenway stoves. All ICS in this project will replace existing traditional wood stoves; therefore, no greenfield installations are included. Further, the project activity will	Baseline survey/manufacture specifications

⁶ [VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves - Verra](#)

⁷ https://verra.org/wp-content/uploads/2022/06/VCS-Standard_v4.3.pdf

⁸ [MethSSC_Guid25ver23.1 \(unfccc.int\)](#)

	utilize only renewable biomass under project scenario. This criterion will be checked from data recorded on the baseline stove used prior to ICS installation	
Project activities shall be implemented in domestic premises or in community-based Kitchen.	The proposed project involves deployment of ICS only in households.	End user Agreement
The project stove shall have specified high- power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass	ICS (Improved Cook Stoves) planned to be installed are cookstoves that have an efficiency of 38% as per the manufacturer's specifications and shall be using woody biomass. The project cookstove shall be using woody biomass which is baseline fuel as well. Further, the project does not involve replacing fossil fuel based baseline stove.	Manufacturer's specification
Both 'Projects' and 'Large Projects' can use the methodology	Estimated average annual emission reductions for each Project Instance may be more than 300,000 tonnes CO ₂ e per year. Therefore, proposed project qualifies the "Large Projects" criteria.	The sample calculation for Emission reductions achieved for a PI has been done below this table.

Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics;	According to Forest policy and tribal development: A study of Maharashtra, in 1989-90, “In the earlier days, fuelwood was easily available at doorsteps of tribals. On the contrary the present situation is that tribal women and children have to spend hours together to collect the fuelwood for everyday use. The hours lost could be used for other employment. The deficit was of fuelwood in Maharashtra was 8.24Mn tonnes in 1976-77. It has increased to 10.78Mn tonnes in 1985-86. This gloomy situation is a result of over-exploitation by the contractors and lack of plantations taken by the forest department for meeting the fuelwood demand.”	Forest policy and tribal development: A study of Maharashtra⁹. The description shows that Non-renewable biomass has been used since 1989.
For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g., shredding and compacting in the	Not applicable. The ICS is introduced as energy efficiency measure to replace baseline stoves and reduce the use of non-renewable biomass for combustion.	Not Applicable

⁹ [Forest policy and tribal development: A study of Maharashtra, 1989](#)

case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.		
The project design shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations	Each ICS in this project will be identified by a unique combination of customer name and geographical location, as well as a serial number. The serial number will be a unique number which will allow for a clear distinction between the stoves. No individual serial number can be repeated within the project, thus ensuring that each stove is counted only once in the proposed project.	Project database and serial no. on ICS's shall be submitted to the VVB. The record keeping system will ensure that each ICS can be traced to project instances to avoid double counting
The project design shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices	The stove manufacturers/wholesale providers/end users shall sign a Declaration on carbon title transfer stating clearly that the PP or an entity authorized by it shall be the sole owner of the VCUs arising from the project.	Carbon title transfer for each user will be recorded and shall be submitted to VVB.

Table 5: Applicability of Methodology

3.3 Project Boundary

As per methodology, the project boundary is the physical, geographical location of all equipment and systems affected by the project activity. As the activity involves distribution of ICS (Improved Cookstove), the project boundary is the physical, geographical location of each household where the ICS (Improved Cookstove) is distributed to replace the traditional stove. The following sources and greenhouse gases have been included in the project boundary.

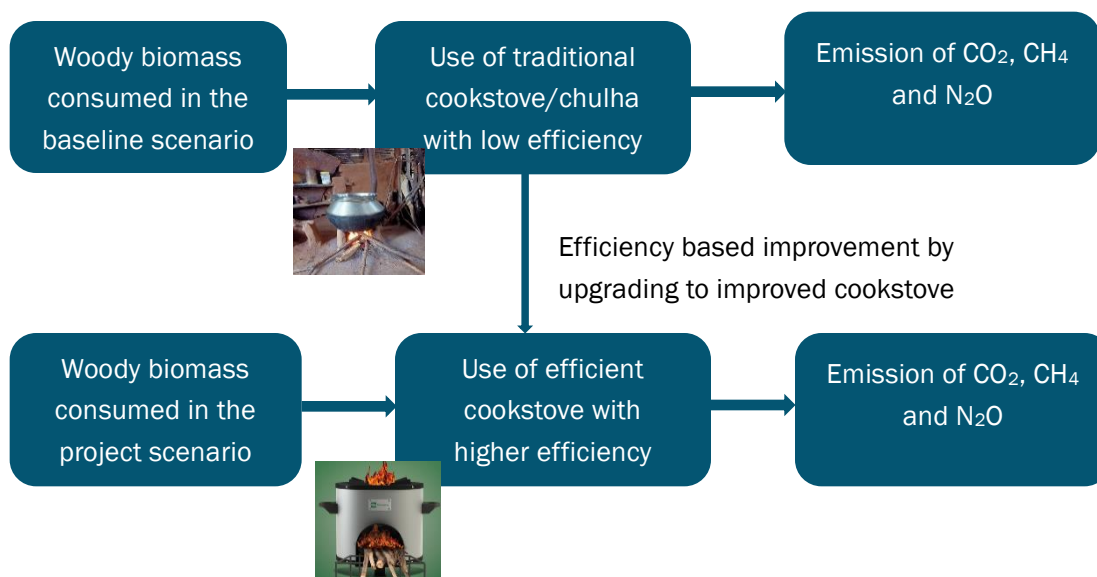


Figure 5: Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable biomass/Fossil fuel	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source
		Other	No	No
Project	Emission from use of non-renewable biomass	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source
		Other	No	No

3.4 Baseline Scenario

The baseline scenario is the continued use of three stone-fired cookstove using non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. For this project the efficiency of the baseline three-stone cookstove has been considered as 10% which is VMR0006 default for efficiency of baseline stove¹⁰.

¹⁰ <https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf>



Figure 6: Baseline Cookstove (Three Stone)

3.5 Additionality

This methodology uses activity method for the demonstration of additionality.

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

The proposed Grouped Project is a voluntary activity initiated by the project proponent. The initiative is solely dependent on VCU revenue for its sustenance and is completely financed by the project developer- AGS Carbon Advisory.

Step 2: Positive List

The Cookstove is disbursed free of cost to the beneficiaries which can be verified by carbon title transfer and end user interviews. Implementation of ICS (Improved Cook stoves) projects pose considerable investment barrier owing to high upfront cost. The efficiency of these improved cookstoves is more than 30% compared to 5%-10% of traditional three stone cookstoves. This makes the procurement and distribution cost of ICS (Improved Cook stoves) much higher, while there is no source of revenue from it apart from carbon credits. The benefit in terms of energy saving is passed on directly to the consumer and the investor can expect returns only from sale of Emission Reductions. As per section 7, step 2 (criteria 1) in VMR0006 *“Methodology for Installation of High Efficiency Firewood Cookstoves”* the project is automatically treated additional if the product is disbursed free of cost to the end user and has no other source of revenue other than the sale of GHG credits.

3.6 Methodology Deviations

The project did not apply any methodology deviations.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The methodology does not account for baseline emissions separately, but instead quantifies net GHG emission reductions achieved by the project. Please refer to Section 4.4.

4.2 Project Emissions

The methodology does not account for project emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with methodology.

4.4 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project. Since ICS shall be replacing the baseline traditional stoves using non-renewable biomass, therefore equations 1 and 2 of the methodology will be applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j}$$

Equation (1)

Where:

i	Indices for the situation where more than one type/model of improved cook stove is introduced to replace three-stone fire
j	Indices for the situation where there is more than one batch of improved cook stove of type i
ER_y	Emission reductions during year y in t CO ₂ e
$ER_{y,i,j}$	Emission reductions by improved cookstove of type i and batch j during year y in tCO ₂ e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{wood\ fuel} \times (EF_{wf,Co2} + EF_{wf,non\ Co2}) \times N_{y,i,j} \times 0.95$$

Equation (2)

Where:

$EF_{wf,CO2}$	CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO ₂ /TJ) ¹¹
$EF_{wf,nonCO2}$	Non-CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO ₂ /TJ)
$N_{y,i,j}$	Number of improved cook stoves of type <i>i</i> and batch <i>j</i> operating during year <i>y</i>
0.95	Discount factor to account for leakage
$B_{y,savings,i,j}$	Quantity of woody biomass that is saved in tonnes per improved cook stove of type <i>i</i> and batch <i>j</i> during year <i>y</i>
$f_{NRB,y}$	Fraction of woody biomass that can be established as non-renewable biomass (fNRB) (Calculated using Tool 30)
$NCV_{wood fuel}$	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)

The quantity of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cook stoves can be estimated by the following Equation:

$$B_{y,savings,i,j} = B_{old} * (1 - \eta_{old} / \eta_{new,i,j}) \quad \text{Equation (3)}$$

Where:

B_{old}	Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove. The value of B_{old} can be sourced from historical data or baseline surveys.
η_{old}	Efficiency of baseline cookstove
$\eta_{new,i,j}$	Efficiency of the improved cook stove type <i>i</i> and batch <i>j</i> determined through water boiling test (WBT)

$$\eta_{new,i,y} = \eta_p * (DF_n)^{y-1} * 0.94 \quad \text{Equation (5)}$$

Where:

η_p	Efficiency of project stove (fraction) at the start of project activity
$(DF_n)^{y-1}$	Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly

¹¹ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

	available literature on relevant industry standards Alternatively default value of 0.99 efficiency loss per year can be considered.
0.94	Adjustment factor to account for uncertainty related to project cookstove efficiency test.)

Where the project households continue to use baseline cookstoves along with improved cookstoves, B_{old} shall be adjusted ex-post based on the percentage of project households found to continue such practice according to Equation 6 of VMR0006 methodology. For such cases, the quantity of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cook stoves shall be calculated using an adjusted value to account for ex-post use of baseline stoves in addition to improved cookstove.

$$B_{old\ adjusted} = B_{old} * (1 - \mu_y) \quad \text{Equation (6)}$$

Where:

$B_{old,adjusted}$	Adjusted B_{old} to account the ex-post usage of firewood in baseline cookstove(s) by project households in addition to improved cookstove (in tonnes per device)
μ_y	Baseline stove usage factor to account for use of baseline cookstoves along with improved cookstoves

For example, considering one ICS/household, the saving is calculated as:

$$B_{y,savings,i,j} = 4.14^{12*} (1 - 0.05) \times [1 - (0.1/0.38)]$$

$$= 2.832 \text{ tonnes}$$

Year (y)	$\eta_{new,y,i,j}$	η_{old}	$B_{y,savings,i,j}$
Year 1	35.72%	0.1	2.832
Year 2	35.36%	0.1	2.821
Year 3	35.01%	0.1	2.810
Year 4	34.66%	0.1	2.798
Year 5	34.31%	0.1	2.787
Year 6	33.97%	0.1	2.775

¹² Based on baseline study conducted by PP. The detailed calculation is provided in the ER ex ante calculation.

Year 7	33.63%	0.1	2.763
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Table 5: Year-wise $B_{y,savings}$ value

Considering one ICS/household, the total number of ICS distributed will be 75,000 hence, the emission reductions is calculated as:

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO2} + EF_{wf,non\ CO2}) \times N_{y,i,j} \times 0.95$$

$$ER_{y,i,j} = 2.832 \times 0.0156 \times 0.913 \times 138.23 \times 75000 \times 1.00 \times 0.95$$

$$= 397,252\ tCO_2$$

Year	$n_{new,i,j}$ (efficiency of cookstove)	$B_{y,savings,i,j}$ (Quantity of wood saved)	Energy Savings per year for all devices (Gwhth)	$ER_{i,j}$ (Emission reduction in tco_2)
1	35.72%	2.832	920	397,252
2	35.36%	2.821	917	375,907
3	35.01%	2.810	913	354,704
4	34.66%	2.798	909	333,645
5	34.31%	2.787	906	312,733
6	33.97%	2.775	902	291,969
7	33.63%	2.763	898	271,356
Total				2,337,567

Table 6: Year-wise estimated emission reductions

Therefore, the annual emission reductions considering total 75,000 ICs are distributed in PI1 are estimated in the table below. Further, as new Project Instances are added the emission reduction table will be updated in the Monitoring Reports.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2022	397,252	0	0	397,252
2023	375,907	0	0	375,907
2024	354,704	0	0	354,704
2025	333,645	0	0	333,645
2026	312,733	0	0	312,733
2027	291,969	0	0	291,969
2028	271,356	0	0	271,356
Total	2,337,567	0	0	2,337,567

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	Bold
Data unit	tonnes/year
Description	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices.
Source of data	Baseline Survey Report
Value applied	4.14
Justification of choice of data or description of measurement methods and procedures applied	This parameter has been fixed ex-ante.

Purpose of Data	Calculation of emission reductions
Comments	Parameter B_{old} once determined shall remain fixed for the entire crediting period.

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass
Source of data	Survey Report
Value applied	0.913
Justification of choice of data or description of measurement methods and procedures applied	The value has been calculated using CDM Tool 30 version 3.0. This parameter has been fixed ex-ante.
Purpose of Data	Calculation of emission reductions
Comments	The report and calculation spreadsheet for f_{NRB} shall be submitted to VVB.

Data / Parameter	$NCV_{wood\ fuel}$
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted or reduced
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions

Comments	The value is fixed ex-ante
Data / Parameter	EF_{wf,CO_2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	Default as per applied approved methodology VMR0006, v1.1
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	Default as per applied approved methodology VMR0006, v1.1
Purpose of Data	Calculation of emission reductions
Comments	The value is fixed ex-ante

Data / Parameter	$EF_{wf,non\ CO_2}$
Data unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	Default as per applied approved methodology VMR0006, v1.1
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	Default as per applied approved methodology VMR0006, v1.1
Purpose of Data	Calculation of emission reductions
Comments	The value is fixed ex-ante

Data / Parameter	η_{old}
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Data unit	Fraction
Description	Efficiency of baseline cookstove
Source of data	Default as per applied approved methodology VMR0006, v1.1
Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney
Purpose of Data	Calculation of emission reductions
Comments	The baseline stove is three stone fired stone using firewood. Hence, efficiency of stove is used 10% which is VMR0006 default. The value is fixed ex-ante

Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of project stove at the start of project activity.
Source of data	Manufacturer's specification
Value applied	0.38
Justification of choice of data or description of measurement methods and procedures applied	This parameter has been fixed ex-ante
Purpose of Data	Calculation of $\eta_{new,y,i,j}$
Comments	The value is fixed ex-ante

Data / Parameter	$\eta_{new,i,y}$
Data unit	Fraction
Description	Efficiency of the improved cookstove type i and batch j implemented as part of the project activity
Source of data	Calculation

Value applied	Year (y)	$\eta_{new,y,i,j}$
	1	35.72%
	2	35.36%
	3	35.01%
	4	34.66%
	5	34.31%
	6	33.97%
	7	33.63%
Justification of choice of data or description of measurement methods and procedures applied	To adopt Option V given in the methodology: "Efficiency of the improved cookstoves to be estimated using equation 5 of applied approved methodology where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored"	
Purpose of Data	Calculation of $\eta_{new,y,i,j}$	
Comments	Fixed ex-ante	

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 95/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and program of activities.
Frequency of monitoring/recording	At least once every two years
Value applied	75,000 (For ex-ante emission reduction calculation)
Monitoring equipment	Monitoring survey

QA/QC procedures to be applied	-
Purpose of data	Calculation of emission reductions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value
Comment	-

Data / Parameter	μ_y
Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y.
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	This parameter should be monitored using one of the following methods: If the baseline cookstoves are decommissioned and no longer used, as determined by the monitoring survey its value is 0 and $\mu_{y, adjusted}$ is equal to $\mu_{y, old}$. If both the improved cookstove and baseline cookstoves are used together then surveys shall be conducted to record the average continued operation of baseline cookstoves in a sample of households. The surveys should be designed to capture the cooking habits and stove usage of households in the region, including quantification of use of baseline cookstoves, by formulating questions and/or collecting evidences to determine the frequency of usage of both the improved cookstoves and baseline cookstoves.
Frequency of monitoring/recording	At least once every two years
Value applied	0.05 (For ex-ante emission reduction calculation)
Monitoring equipment	Monitoring survey
QA/QC procedures to be applied	A 95 /10 confidence / margin of error shall be achieved for the sampling parameter irrespective of annual/ biennial monitoring frequency as per para 22 of Standard:

	Sampling and surveys for CDM project activities and programmes of activities, Version 09.0. In the case the desired precision is not met, lower bound will be applied.
Purpose of data	Calculation of emission reductions
Calculation method	Not applicable
Comments	-

Data / Parameter	Date of commissioning of batch j
Data unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch.
Source of data	Project database
Description of measurement methods and procedures to be applied	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch.
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch
Value applied	06/06/2022
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Not applicable
Purpose of data	Calculation of emission reductions
Calculation method	Not applicable
Comments	-

Data / Parameter	Life Span
Data unit	Number of years
Description	Project promoters to state the operating lifetime of project device for projects opting Equation 9 for determining project stove efficiency.
Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	The cookstoves will be manufactured to match the fixed design specification. This will ensure that each stove that is built at individual end user household measures exactly same as the dimensions specified by the manufacturer. Post construction, training will be provided to end users on use, care, and upkeep of these stoves. PP will conduct periodic audits and surveillance of the stoves that will be distributed under the project activity to ensure their proper functioning throughout the project lifetime. This along with spot audits and after installation maintenance services, will ensure that the project stoves continue to work at efficiencies as specified by the manufacturer.
Frequency of monitoring/recording	Once at the time of project stove installation
Value applied	6 years
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of emission reductions
Calculation method	-
Comments	-

5.3 Monitoring Plan

Description of the monitoring plan

The following are the monitoring requirements as defined in methodology VMR0006.; Version V1.

The Monitoring Plan applied in this project involves several key elements that ensure that the PP have high-quality, unbiased, and reliable information regarding the performance of the project in

terms of implementation and outcomes, and for the purposes of calculating Verified Carbon Units (VCUs) following VMR 0006 version 1.1 on the basis of the amount of non-renewable biomass saved by the ICS in the project activity. The key elements are the following:

- **Data collection procedures**

Project implementer will collect/receive the necessary information requested in the Registration process from the user. Means of collecting this information may be through a physical Registration Card filled by project Imp staff, retailers, end-users, or partner organization's staff, or through the use of ICTs or SMS. Project Implementers' staff shall double check the accuracy of information provided, and request for field staff additional clarifications if needed.

- **Distribution and Monitoring Database**

Project implementer trained staff will input the data in the database either manually (if data collected from physical Registration Card) or this will be automatically input if data was collected using ICTs or SMS. Project implementer staff shall double check the information included on the database and check for duplications. Any duplicate information shall be investigated, and errors corrected or excluded from the database if it is a true duplicate entry.

- **Spot Checking of ICS (ongoing)**

Project implementer field staff will randomly select units included in the database and visit or contact the stove users to cross-check the information on the database with the factual evidence in the field. Any inconsistencies found (e.g., change in the address of a user) will be updated on the database, and in the case, ICS are found to be no longer in use, they will be clearly marked as such and excluded from emission reductions calculations.

- **Sample Plan for the Monitoring Survey**

SAMPLING PLAN

S. No	Information head	Coverage
1	Sampling Objectives	The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period. As per Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities version 9.0, 90/10 confidence/precision to be adopted for small-scale project and 95/10 for large-scale project. Given that the size of the project is under the category of large projects, hence 95/10 confidence/precision shall be adopted for all parameters. 1. $N_{y,i,j}$ Number of project devices of type i and

		batch j operating during year y (Annual/biennial) 2. μ_y Adjustment to account for any continued use of pre-project devices during the year y
2	Target Population	The target population are all households in the project database which are using fuelwood in ICS distributed under the project for cooking.
3.	Sampling Frame.	The sample frame refers to all the information sources on the Database. The SMS/ICT system is the primary mechanism for data collection for newly distributed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the project. The units in the sampling frame shall meet the following criteria: - All units will have a unique identification no. - All units can be easily traced –contact information, geographical location or other relevant information of beneficiary is present - Every element of the population is present only once in the frame - No elements from outside the population of interest are present in the frame Where information miss-match is observed, conservative assumptions would be applied. Sampling frame for proportion of ICS still in operation ($N_{y,j}$) The first step is to identify the Primary Sampling Units. Factors such as model of the ICS do not effect parameter $N_{y,j}$. Therefore Primary sampling unit in case of parameters $N_{y,j}$ is independent of model of stove.

4	Sample Method	The sampling method for the monitored parameters $N_{y,i,j}$, and $u_{y,i,j}$ is Simple Random Sampling and samples will be randomly selected from the primary sampling units as illustrated above. To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the project proponent. To determine the parameters, sampling will involve the following approaches (outcome inbrackets): $N_{y,i,j}$: Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No) $\mu_{y,i,j}$: Pre project device only is in use then fraction to be used to calculate total number, however if pre project device is used along with project ICS, proportion of usage of each will be determined by cooking habits evaluated by survey questionnaire during the monitoring period. Using the formulas below, the project proponent will randomly sample the required number of ICS from the primary sampling units.
5	Desired Precision / Expected Variance and Sample Size.	To estimate the number of premises to be surveyed with a 10% margin of error at desired confidence level of 95%, the optimal sample size n is given by: 1. For proportional parameters

		$n \geq \frac{1.96^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.96^2 p(1-p)}$ Here, n = Sample size N = Total number of Cookstoves (ICS) distributed p = Our expected proportion 1.96 = Represents the 95% confidence required (In the case of 90% confidence, 1.645 shall be used) 0.1 = Represents the 10% relative precision (0.1x0.5 = 0.05 =5% points either side of p) The above equation has two unknowns- n and p. However, using information from a pilot study, a possible value of p, can be used for estimating n. 2. For mean-based parameters $n \geq \frac{1.96^2 NV}{(N-1) \times 0.1^2 + 1.96^2 V}$ Here, n = Sample size N = Total number of Cookstoves (ICS) distributed V = (SD/mean)² SD = Standard deviation 1.96 = Represents the 95% confidence required (In the case of 90% confidence, 1.645 shall be used) 0.1 = Represents the 10% relative precision (0.1x0.5 = 0.05 =5% points either side of p) Requirements of Guideline for Sampling and surveys for CDM project activities and program of activities, version 4.0 shall be followed.
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6	Minimum Sample Size	If the sample size calculation returns a value of less than 30 samples, a minimum sample size of 30 shall be chosen when the parameter of interest is a proportion.
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Table 7: Sampling Plan

- **Quality Assurance/Quality Control**

The project proponent will apply measures to ensure the required confidence/precision for each sampled parameter is met, allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The choice of measure applied to each parameter will depend on the cost of each data collection approach and logistics required. The project proponent may choose to stop monitoring a particular parameter once the required level of confidence/precision has been reached, as long as the calculated minimum number of samples has been achieved.. The project proponent will determine the most effective measure for each parameter from the following list:

- **Oversampling:** Randomly draw a sample more than the calculated number (say 20%) and collect data from each
- **Buffer Group:** Randomly draw additional samples (say 20%) and collect data from only for minimum numbers of ICS as per sample size calculation. If this would not result in the required sample size data would be collected from the additional ICS that were selected in the sample.
- If precision required is not achieved by reliability check, use the lower bound or upper bound of estimates of the parameter.
- Addressing non-sampling errors, such as non-response or bias from interviewer.

The monitoring and sampling plan has the following procedures in place to ensure good quality data. The project proponent will ensure that field personnel have reviewed, understand and have agreed to follow the monitoring plan procedures, including provisions for maximizing response rates, documenting out-of-population cases, refusals and other sources of non-response. A quality control and assurance strategy will be audited and documented. The monitoring plan will explain how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household will be chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires will be used. The calculation of the sample size will be carried out using estimates for parameter proportions, mean values and standard deviations, as the actual characteristics of the population/sampling frame are unknown. In order to ensure the quality of the sampling results, the project proponent can draw on the provisions for reliability calculations as provided by the Guidelines for Sampling and Surveys in CDM Project Activities and Program of Activities (version 04). In the event that the sampling results do not fulfil the required level of confidence and precision, the project proponent can undertake additional samples. If the reliability is still not sufficient after additional samples or other measures, the sampling may be repeated with an increased sample size. Alternatively, the project proponent may choose to apply the lower bound or higher bound according to the more conservative approach.

- **Monitoring Reporting**

The PP will assess all monitoring data and produce one or two monitoring reports for the project for the VVB to verify corresponding to the preceding monitoring period the project. This report will present the data relating to the emission reductions generated by the project at the time of the monitoring period.

A. Personnel Training

Separate training will be given to deal adequately with surveying techniques and reduce errors. The training will include survey techniques, efficiency of cookstove, frequency of survey that needs to be done. And, evidence (attendance sheet, power point presentations) in support of the training should be documented. Quarterly training will be given with refresher training before the monitoring activity.

B. Roles and Responsibilities of Project Proponent

- Development of VCS project Description using latest template.
- Registration of the project with VERRA.
- PP itself shall officially communicate with VERRA for project registration
- Providing improved cookstove (ICS) to replace traditional cookstove.
- Getting the project validated/verified by a Designated Operational Entity
- Carrying out annual surveys and monitoring
- Maintaining the project database and other documentation required by VERRA
- Preparing the Monitoring Reports for the Pls.

C. Organizational Structure

Person	Role
PP database administrator	The database administrator is responsible for updating and maintaining all electronic databases. Required competencies include experience with data management systems (e.g. Excel, STATA, or SPSS), minimum 2 years working experience in a similar field, and at minimum a Bachelor's degree from an institution of higher education.
Monitoring team	The monitoring team will be assigned by the PP to conduct the user interviews and check the operational status of Cookstoves (ICS) in Households during the periodic sampling and reports the results to the database administrator. The skills and experience required for the data collection activities include: • Experience conducting surveys/tests

	• Experience conducting door-to-door surveys • Local language skills (especially important for input to questionnaire design and interviewing of end users) • English language skills • Cultural awareness • Numerical proficiency • Data entry skills
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Table 8: Organization Structure

- Quality Assurance and Quality Check

PP will undertake the following strategies, tailoring the specific approach to the local circumstances:

- 1) Ensuring end user awareness. At the time of sale, the CEP customer is made aware that they are required to participate in monitoring activities. This will be via training sales personnel to explain the importance of monitoring to each customer, and during regularly scheduled microfinance group meetings for end-users.
- 2) Questionnaire design. The design of the questionnaire will ensure that the questions are non-intrusive and easy to understand for both the interviewee and interviewer.
- 3) Drawing on local knowledge. The local contractors to be hired by the PP in each region will play an important role in tailoring the approach to suit local circumstances. For example, in some instances, it may be essential for a local person to conduct the interview in order to obtain accurate results.
- 4) Quality of contractors. Any third parties hired by the PP to carry out sampling will be required to demonstrate a high level of cultural awareness, local language skills and appropriate experience with data entry and data management. Training will also be provided on how to deal with non-responses, refusals and other problems should these occur.